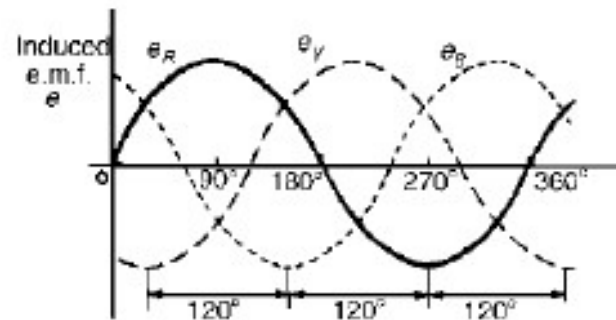
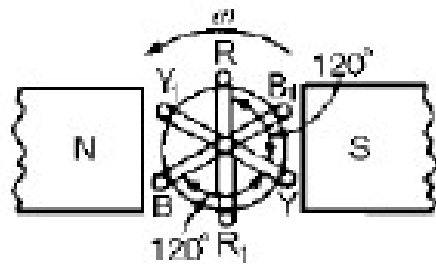


**LECTURE –VII**  
**THREE-PHASE SUPPLY SYSTEMS**

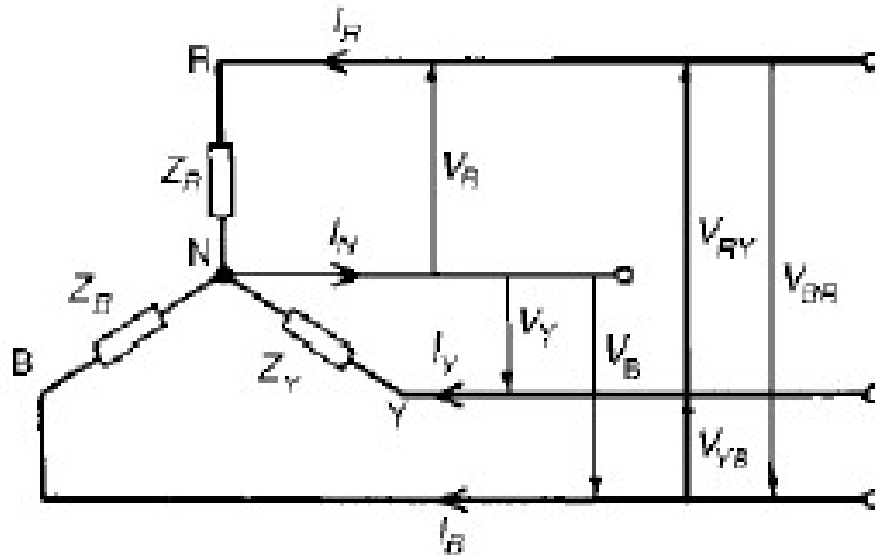
**A three-phase supply** is generated when three coils are placed  $120^\circ$  apart and the whole rotated in a uniform magnetic field as shown in Figure . The result is three independent supplies of equal voltages which are each displaced by  $120^\circ$  from each other as shown in Figure .

- (i) The convention adopted to identify each of the phase voltages is: R-red, Y-yellow, and B-blue, as shown in Figure



The **phase-sequence** is given by the sequence in which the conductors pass the point initially taken by the red conductor. The national standard phase sequence is R, Y, B.

## Star Connected Loads



A **star-connected load** is shown in Figure where the three line conductors are each connected to a load and the outlets from the loads are joined together at  $N$  to form what is termed the **neutral point** or the **star point**.

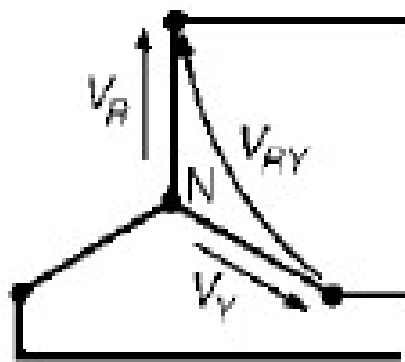
The voltages,  $V_R$ ,  $V_Y$  and  $V_B$  are called **phase voltages** or line to neutral voltages. Phase voltages are generally denoted by  $V_P$

For a balanced system:  $I_R = I_Y = I_B$ ,  $V_R = V_Y = V_B$

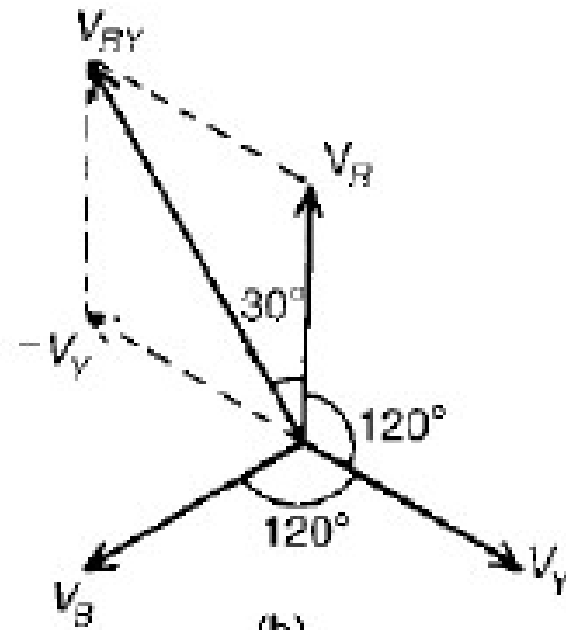
$$V_{RY} = V_{YB} = V_{BR}, \quad Z_R = Z_Y = Z_B$$

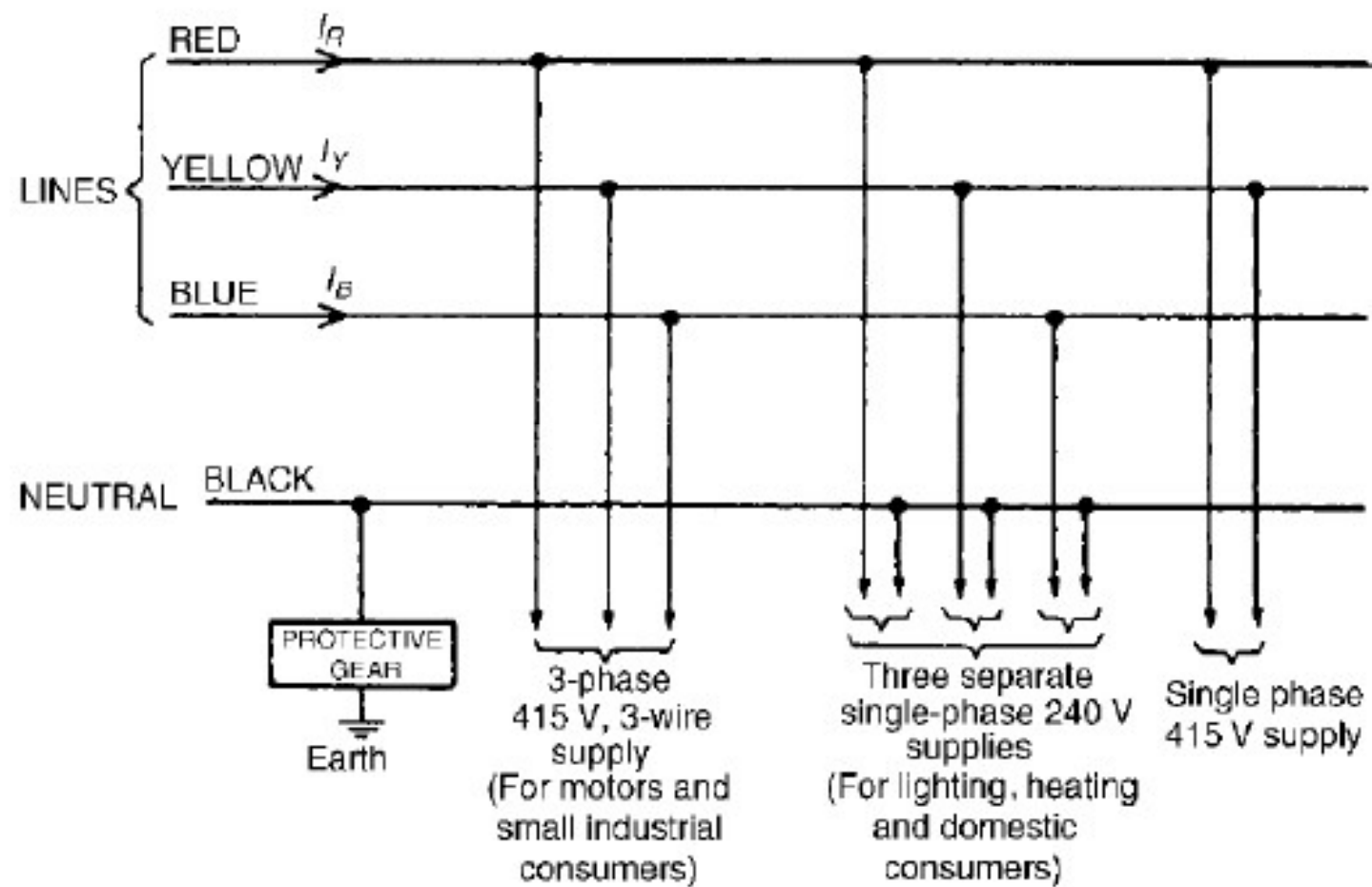
and the current in the neutral conductor,  $I_N = 0$

When a star connected system is balanced, then the neutral conductor is unnecessary and is often omitted.



$$V_{RY} = V_R - V_Y$$





**Problem** Three loads, each of resistance  $30\ \Omega$ , are connected in star to a  $415\ \text{V}$ , 3-phase supply. Determine (a) the system phase voltage, (b) the phase current and (c) the line current.

A '415 V, 3-phase supply' means that  $415\ \text{V}$  is the line voltage,  $V_L$

(a) For a star connection,  $V_L = \sqrt{3}V_P$

$$\text{Hence phase voltage, } V_P = \frac{V_L}{\sqrt{3}} = \frac{415}{\sqrt{3}} = \mathbf{239.6\ \text{V or } 240\ \text{V}}$$

correct to 3 significant figures

(b) Phase current,  $I_P = \frac{V_P}{R_P} = \frac{240}{30} = \mathbf{8\ \text{A}}$

(c) For a star connection,  $I_P = I_L$   
Hence the line current,  $I_L = \mathbf{8\ \text{A}}$

**Problem** A star-connected load consists of three identical coils each of resistance  $30\ \Omega$  and inductance  $127.3\text{ mH}$ . If the line current is  $5.08\text{ A}$ , calculate the line voltage if the supply frequency is  $50\text{ Hz}$

Inductive reactance  $X_L = 2\pi fL = 2\pi(50)(127.3 \times 10^{-3}) = 40\ \Omega$

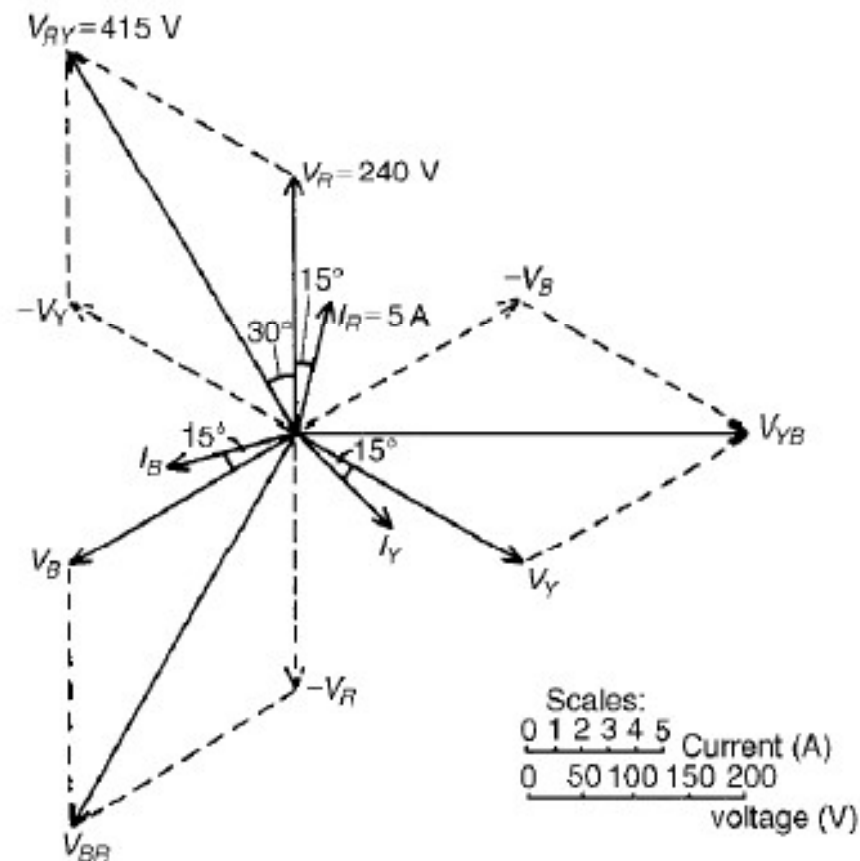
Impedance of each phase  $Z_p = \sqrt{R^2 + X_L^2} = \sqrt{30^2 + 40^2} = 50\ \Omega$

For a star connection  $I_L = I_p = \frac{V_p}{Z_p}$

Hence phase voltage  $V_p = I_p Z_p = (5.08)(50) = 254\text{ V}$

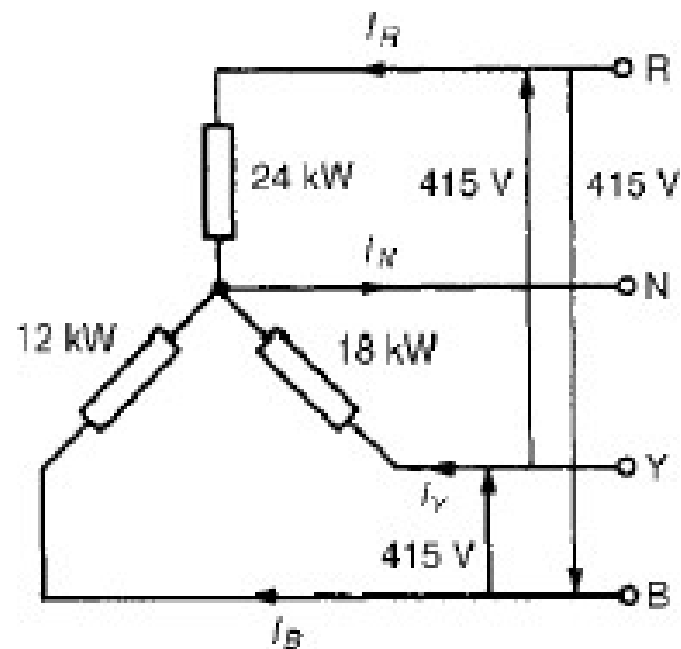
Line voltage  $V_L = \sqrt{3}V_p = \sqrt{3}(254) = \mathbf{440\text{ V}}$

**Problem** A balanced, three-wire, star-connected, 3-phase load has a phase voltage of 240 V, a line current of 5 A and a lagging power factor of 0.966. Draw the complete phasor diagram.





**Problem** A 415 V, 3-phase, 4 wire, star-connected system supplies three resistive loads as shown in Figure Determine (a) the current in each line and (b) the current in the neutral conductor.



(a) For a star-connected system  $V_L = \sqrt{3}V_p$

$$\text{Hence } V_p = \frac{V_L}{\sqrt{3}} = \frac{415}{\sqrt{3}} = 240 \text{ V}$$

Since current  $I = \frac{\text{Power } P}{\text{Voltage } V}$  for a resistive load

$$\text{then } I_R = \frac{P_R}{V_R} = \frac{24\,000}{240} = \mathbf{100 \text{ A}}$$

$$I_Y = \frac{P_Y}{V_Y} = \frac{18\,000}{240} = \mathbf{75 \text{ A}}$$

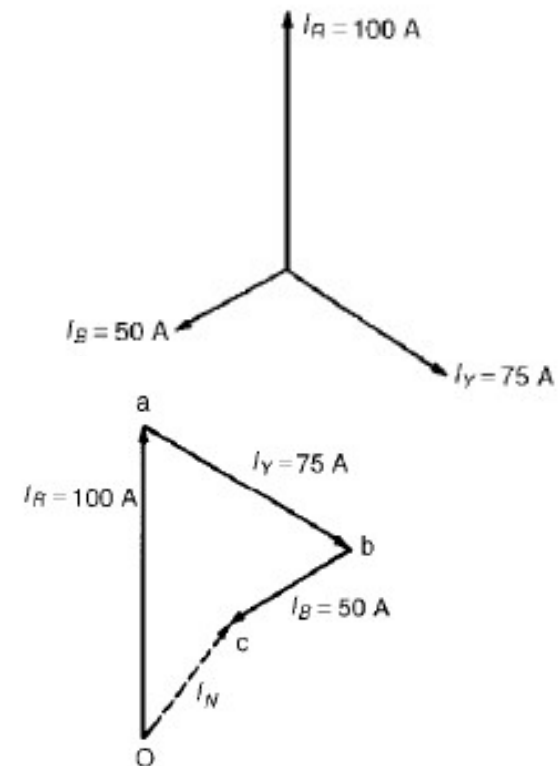
$$\text{and } I_B = \frac{P_B}{V_B} = \frac{12\,000}{240} = \mathbf{50 \text{ A}}$$

The three line currents are shown in the phasor diagram of Figure . Since each load is resistive the currents are in phase with the phase voltages and are hence mutually displaced by  $120^\circ$ . The current in the neutral conductor is given by:

$$I_N = I_R + I_Y + I_B \text{ phasorially.}$$

Figure shows the three line currents added phasorially.  $oa$  represents  $I_R$  in magnitude and direction. From the nose of  $oa$ ,  $ab$  is drawn representing  $I_Y$  in magnitude and direction. From the nose of  $ab$ ,  $bc$  is drawn representing  $I_B$  in magnitude and direction.  $oc$  represents the resultant,  $I_N$ .

By measurement,  $I_N = 43 \text{ A}$



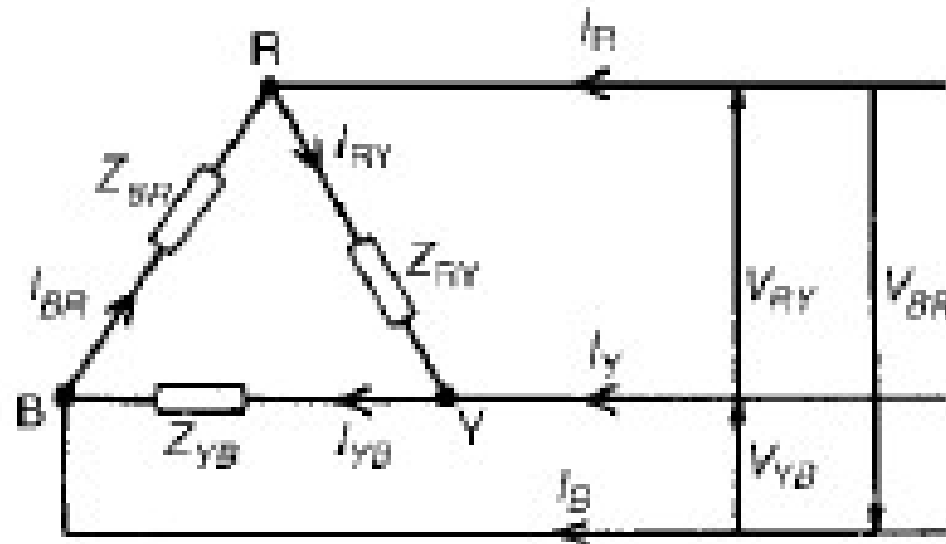
Alternatively, by calculation, considering  $I_R$  at  $90^\circ$ ,  $I_B$  at  $210^\circ$  and  $I_Y$  at  $330^\circ$ :

$$\begin{aligned}\text{Total horizontal component} &= 100 \cos 90^\circ + 75 \cos 330^\circ + 50 \cos 210^\circ \\ &= 21.65\end{aligned}$$

$$\begin{aligned}\text{Total vertical component} &= 100 \sin 90^\circ + 75 \sin 330^\circ + 50 \sin 210^\circ \\ &= 37.50\end{aligned}$$

$$\text{Hence magnitude of } I_N = \sqrt{(21.65^2 + 37.50^2)} = \mathbf{43.3 \text{ A}}$$

# Delta Connection

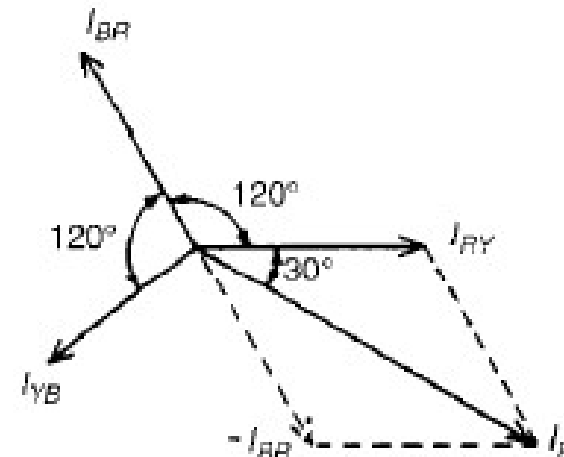


- (i) A **delta (or mesh) connected load** is shown in Figure        where the end of one load is connected to the start of the next load.
- (ii) From Figure        it can be seen that the line voltages  $V_{RY}$ ,  $V_{YB}$  and  $V_{BR}$  are the respective phase voltages, i.e. for a delta connection:

$$V_L = V_p$$

- (iii) Using Kirchhoff's current law in Figure        ,  $I_R = I_{RY} - I_{BR} = I_{RY} + (-I_{BR})$ . From the phasor diagram shown in Figure        , by trigonometry or by measurement,  $I_R = \sqrt{3}I_{RY}$ , i.e. for a delta connection:

$$I_L = \sqrt{3}I_p$$



**Problem** Three identical coils each of resistance  $30\ \Omega$  and inductance  $127.3\text{ mH}$  are connected in delta to a  $440\text{ V}$ ,  $50\text{ Hz}$ , 3-phase supply. Determine (a) the phase current, and (b) the line current.

Phase impedance,  $Z_p = 50\ \Omega$  (from problem 2) and for a delta connection,  $V_p = V_L$

(a) Phase current, 
$$I_p = \frac{V_p}{Z_p} = \frac{V_L}{Z_p} = \frac{440}{50} = \mathbf{8.8\text{ A}}$$

(b) For a delta connection,  $I_L = \sqrt{3}I_p = \sqrt{3}(8.8) = \mathbf{15.24\text{ A}}$

*Thus when the load is connected in delta, three times the line current is taken from the supply than is taken if connected in star.*

**Problem** Three identical capacitors are connected in delta to a 415 V, 50 Hz, 3-phase supply. If the line current is 15 A, determine the capacitance of each of the capacitors.

For a delta connection  $I_L = \sqrt{3}I_p$

Hence phase current  $I_p = \frac{I_L}{\sqrt{3}} = \frac{15}{\sqrt{3}} = 8.66 \text{ A}$

Capacitive reactance per phase,  $X_C = \frac{V_p}{I_p} = \frac{V_L}{I_p}$  (since for a delta connection  $V_L = V_p$ )

Hence  $X_C = \frac{415}{8.66} = 47.92 \text{ } \Omega$

$X_C = \frac{1}{2\pi f C}$ , from which capacitance,  $C = \frac{1}{2\pi f X_C} = \frac{1}{2\pi(50)(47.92)} \text{ F}$   
 $= 66.43 \text{ } \mu\text{F}$

**Problem** Three coils each having resistance  $3\ \Omega$  and inductive reactance  $4\ \Omega$  are connected (i) in star and (ii) in delta to a  $415\text{ V}$ , 3-phase supply. Calculate for each connection (a) the line and phase voltages and (b) the phase and line currents.



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